

Software Architecture Fundamentals & Quality Attributes

CS673 Software Engineering

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1. I have chosen Microsoft Outlook for this assignment.
2. I believe that Microsoft employed a hybrid approach incorporating elements of Client-Server Model, Microservices, and Layered Architecture.
 - a. Client-Server Model: Outlook clients (desktop, mobile, web) communicate with backend servers for data synchronization, email delivery, and other services.
 - b. Microservices: Certain functionalities, like spam filtering, calendar scheduling, or search indexing, likely operate as independent microservices. This allows for independent scaling and updates.
 - c. Layered Architecture: The system likely employs a layered structure, separating the presentation layer (UI), application layer (business logic), and data layer (database).
3. I have chosen to discuss how the architecture supports scalability, reliability, and maintainability.
 - a. Scalability: The client-server model allows for horizontal scaling by adding more servers to handle increased user load. Microservices architecture further enhances scalability by allowing individual services to scale independently based on their specific needs. For example, the email storage service can be scaled independently of the calendar service.
 - b. Reliability: Redundancy and failsafe mechanisms are crucial. Multiple servers are likely deployed for each service, ensuring that if one fails, others can take over. Microservices architecture enhances fault isolation. If one service fails, it is less likely to bring down the entire system. Data replication and backups are also essential for reliability.
 - c. Maintainability: The layered architecture promotes modularity, making it easier to modify or update specific components without affecting others. Microservices architecture enhances maintainability by enabling independent deployments and updates. Smaller, focused services are easier to understand and manage than a monolithic codebase.
4. There are certain trade-offs associated with each of the attributes.
 - a. Scalability: Scaling requires investment in infrastructure and potentially more complex deployment and management processes. Scaling data storage and processing can sometimes conflict with maintaining strong data consistency. Data consistency across distributed microservices becomes a challenge. Outlook likely employs eventual consistency models in some areas to improve performance and scalability, accepting that minor delays in data synchronization might occur.
 - b. Reliability: Implementing redundancy adds complexity and cost. Building highly reliable systems with redundancy and failsafe mechanisms increases infrastructure costs. Microsoft had to balance the desired reliability level with the cost of implementation and maintenance.
 - c. Maintainability: While microservices enhance long-term maintainability, they can initially increase complexity due to distributed systems, inter-service communication, and the need for robust testing strategies. Managing dependencies between microservices can also be challenging. Outlook's architecture likely reflects a trade-off between embracing microservices for future flexibility while managing the initial complexity.